

Lewis on fission; Olson on Animalism

PHIL UA-78 Metaphysics

Prof. Cian Dorr

9th September 2026

Coincidence, temporary and permanent

Can two material objects be in the same place at the same time?

When people first hear this question, they start imagining weird science fiction or fantasy scenarios where certain objects can pass through one another, like ghosts.

But there is a strong argument that the answer is yes. Consider a statue made of a lump of clay, which will later be enlarged by the addition of more clay.

1. The statue is now in exactly the same place as the lump, and both are material objects.
2. The statue will later be larger than the lump.
3. The lump will never be larger than itself.
4. So, the statue is not identical to the lump. (By Leibniz's Law)
5. So, two material objects are in exactly the same place at the same time.

Recap: Spatiotemporalism about material objects

[A person is] a streak through the manifold of space-time, a whole composed of stages located at various times and places. (Lewis, 'Paradoxes of Time Travel')

We can read this as endorsing:

Spatiotemporalism

Every material object is a region of spacetime.

Recap: Spatiotemporalism about material objects

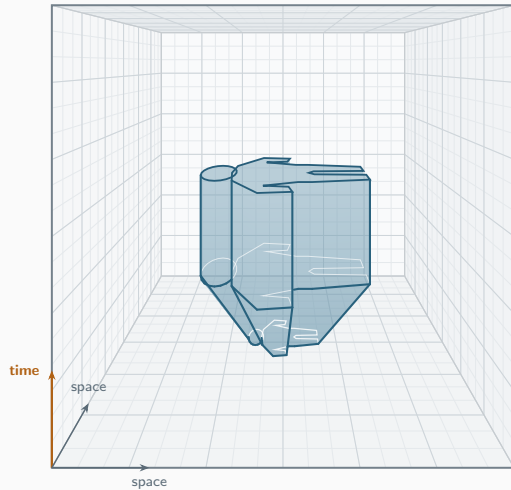
[A person is] a streak through the manifold of space-time, a whole composed of stages located at various times and places. (Lewis, 'Paradoxes of Time Travel')

We can read this as endorsing:

Spatiotemporalism

Every material object is a region of spacetime.

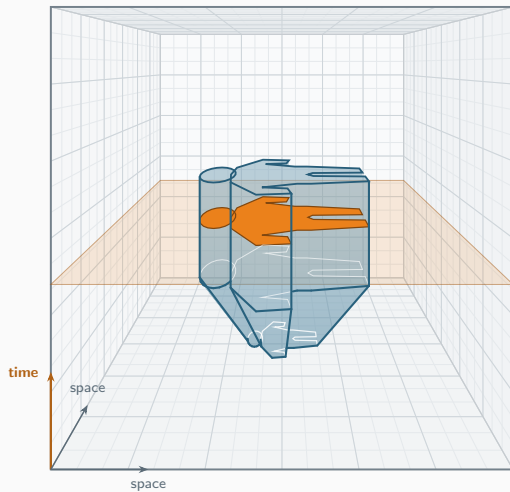
- ▶ Should Spatiotemporalists also endorse the converse, 'Every region of spacetime is a material object'? (Or at least, 'Every region that overlaps a time is a material object at that time'?)
- ▶ If the physics provides a principled contrast between 'empty' and 'filled' points (where more than negligibly many points are empty), we could say that being material at t requires including only 'filled' points at t . But my read is that this seems unlikely.

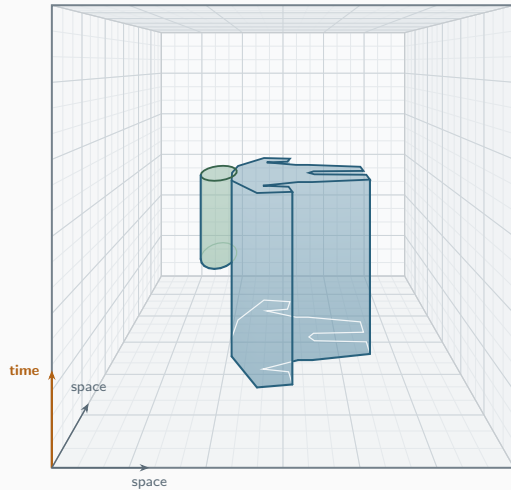


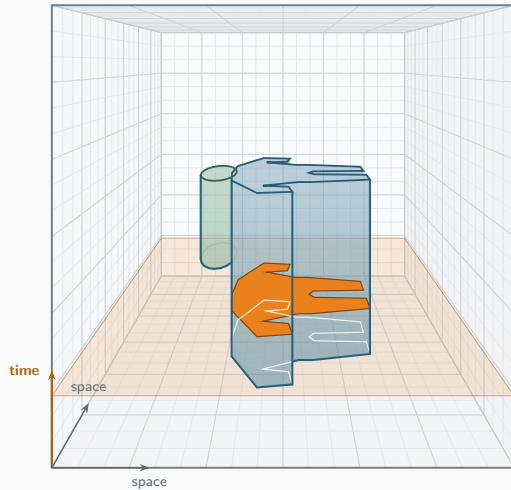
Recall that regions x and y **coincide at** t iff $x \cap t = y \cap t \neq \emptyset$.

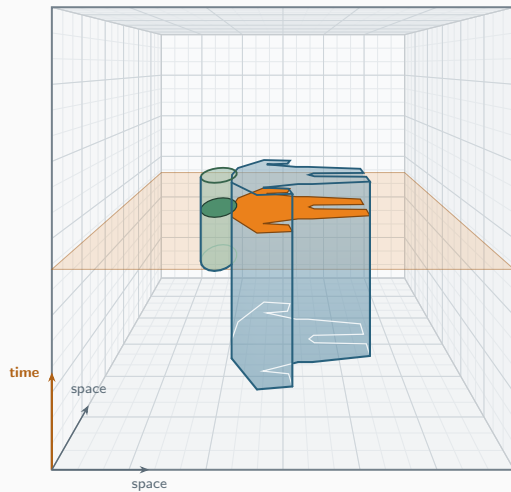
Objects that coincide at t are in the same place at t . So, spatiotemporalism can easily accommodate cases where two objects are in the same place at the same time.

A stage: the object's intersection with one time









However, spatiotemporalism *cannot* allow for the possibility of two distinct material objects coinciding at *every* time at which either of them concretely exists.

Since every spacetime point belongs to some time, if $x \cap t = y \cap t$ for all t , then $x = y$.

Goliath and Lumpl

Alan Gibbard ('Contingent Identity') describes a case which seems to be like this. In Gibbard's case, a clay statue 'Goliath' and a lump of clay 'Lumpl' are formed simultaneously by merging two prefabricated halves, and later destroyed simultaneously by dissolution in acid.

- ▶ I think the case that Lumpl didn't exist before Goliath is clearer if we make it bronze rather than clay, formed by melting copper and tin in a mould.

We can give various Leibniz's Law arguments that Goliath is not identical to Lumpl:

- ▶ Goliath *could* have become larger than Lumpl later (by the attachment of some additional bits of clay parts), whereas Lumpl, of course, could not.
- ▶ Goliath *would* have become larger than Lumpl under those circumstances, whereas Lumpl would not.
- ▶ Lumpl is (let's suppose) well-made (smooth, well-kneaded... or good copper/tin ratio), whereas Goliath is not (awkwardly modelled nose,...).

FYI, Lewis is aware of this objection and has a response to it, based on positing extensive *context-sensitivity* in the words that feature in the LL arguments.

- ▶ E.g.: 'well-made' can mean 'well-made considered as a piece of bronze' or 'well-made considered as a statue'. Although the names 'Goliath' and 'Lumpl' refer to the same object, which one we choose affects how it's natural to resolve the context-sensitivity of other words.
- ▶ How does this work for modals?

An alternative to Spatiotemporalism

A quite different response would be to reject the Spatiotemporalist view of objects as *collections* of points of spacetime, in favour of the following:

Property Spatiotemporalism

Every material object is a *property* of points of spacetime.

The key contrast between *properties* and *collections* is that two distinct collections can't have the same elements, whereas two distinct properties *can* have the same instances.

- Note that this leads to an *even more enormous* number of material objects coincident with any given material object than we already had given Spatiotemporalism.

Puzzles of fission (again)

A case of *personal fission* is one where a previously more or less normal person undergoes some process such that at the end, we seem to have two people, for each of whom there is *some* temptation to think that they did not just recently come into existence or come to be a person, but lived an earlier life as a person—the very same earlier life!

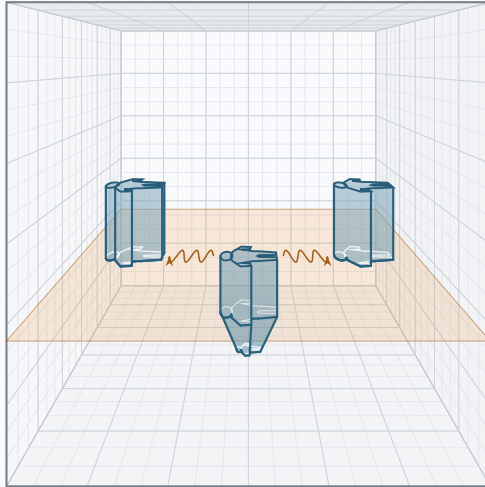
Some worms have the ability to regrow their bodies such that, if chopped in half, we end up with two worms. One could imagine a serum that gives people a similar ability. The literature focuses on a case of brain-bisection plus transplantation.

Double teletransportation and the branch-line case

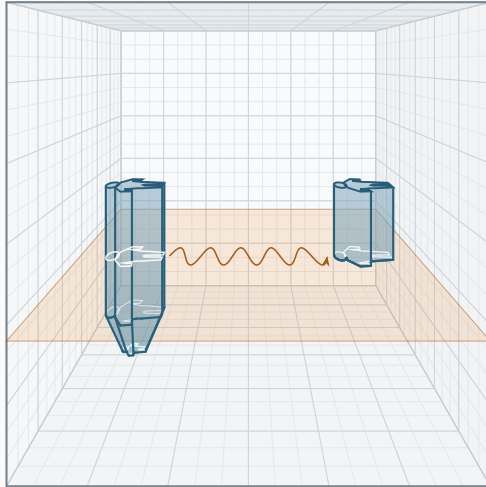
Two other cases from the literature:

- ▶ Double teletransportation. Like regular teletransportation except that Replicas are created in two different places.
- ▶ Parfit's branch-line case: like regular teletransportation except that the scanning process does not involve any destruction.

Double teletransportation



The branch-line case



Mapping out the possible views of a fission case

(Say that something exists as a person at t iff (a) it is concrete at t , and (b) it is a person at t .)

(i) Nothing that exists as a person after the fission existed as a person before the fission. (Parfit on brain-bisection and double teletransportation)

(ii) Everything that exists as a person after the fission already existed as a person before the fission. (Lewis on all our cases)

(iii) Some but not all of the things that exist as people after the fission already existed as people before the fission. (Parfit on branch-line teletransportation)

Type (ii) ('No new person') views

The problem for type (ii) views is that there would seem to be *exactly one thing* that exists as a person before the fission, and *exactly two* that exist as people afterwards. They'll have to deny at least one of these claims.

In principle, you could imagine a view where, despite appearances, there is only one person after the fission—a very unusually shaped, spatially scattered person with four arms, two heads, and a severe case of split personality. This has not been popular.

More widely discussed is Lewis's view where, despite appearances, there are already two (or more!) people before the fission—two unusually similar people in exactly the same place, with exactly the same shape, made of exactly the same matter, with exactly the same mental properties. . .

Lewis adopts a very surprising view of such cases! According to him, there were **two people all along**. They coincide before the teletransportation and go their separate ways afterwards.

This also happens in more symmetric 'fission' cases such as brain-bisections (where both halves retain enough mental similarity to the original). (And in 'fusion' cases, which are the same thing in reverse.)

Proper names in use before the fission were *ambiguous*; presumably this also applies to personal pronouns like 'I'.

Lewis's analysis of 'person'

x is a person iff x is a *maximal R -interrelated aggregate of person-stages*.

The ' R -relation' is a relation of *mental continuity and connectedness* between stages.
(Note that this assumes Lewis's view that stages have mental properties!)

An 'aggregate' is just a union/join.

A *maximal F* is an F that isn't part of any other F .

The counting objection to Lewis on fission and fusion

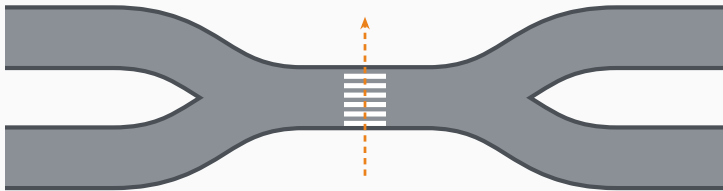
Lewis's theory says that there's a person that goes to Mars, and a person that doesn't go to Mars, that initially coincide but are not numerically identical. Given the standard view about the connection between counting-words and identity (which we have been presupposing so far), it follows that there are two people in the room before the fission. This sounds obviously wrong!

Lewis's response is to **reject the standard theory of counting**. Instead, he thinks that expressions like 'exactly one', 'at least two' etc. are context-sensitive, and sometimes mean things that can be spelled out using relations other than identity.

- ▶ E.g.: 'There is now at most one person in this room' can mean 'Every person in this room coincides now with every person in this room'.
- ▶ This is **weird**! 'There is at most one person in this room, and there is a person who will be rich in the room, and there is a person who will never be rich in the room' *sounds* inconsistent—but on Lewis's view, it has a possibly-true reading!

Lewis's argument for the revisionary theory of counting

It may seem far-fetched to claim that we ever count persons other than by identity simpliciter. But we sometimes *do* count otherwise. If an infirm man wishes to know how many roads he must cross to reach his destination, I will count by identity-along-his-path rather than by identity. By crossing the Chester A. Arthur Parkway and Route 137 at the brief stretch where they have merged, he can cross both by crossing only one road. Yet these two roads are certainly not identical. (Lewis, S&I, 27)



What else might one say about the roads?

An alternative account of this case would explain it in terms of context-sensitivity in 'road' rather than context-sensitivity in 'one'. Maybe, in the context where I say 'You only have to cross one road', 'road' expresses a property with just one instance in the domain of your quantifier (perhaps a pretty short "stretch of road").

- An argument for this: the helpful answer to 'How many roads must I cross?' would be 'Two' even if it's really crossing the Chester A. Arthur Parkway both times (because it makes a loop).